SKILLRANK: INTERNSHIP PORTAL AND CANDIDATE RANKING SYSTEM

FINAL YEAR PROJECT REPORT SUBMITTED IN PARTIAL FULLFILMENT

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Chapter 1:INTRODUCTION

1.1 Background of the study

In the current job market, internships play a crucial role in bridging the gap between academic learning and professional experience. However, the traditional process of internship recruitment is often manual, time-consuming, and inefficient. Employers struggle to identify suitable candidates from a large pool of applicants, while students face challenges in finding relevant opportunities that match their skills and interests.

With the rapid growth of digital platforms, there is a need for an automated system that can streamline internship applications, match candidates based on skills, and provide transparent ranking to help employers make faster and better hiring decisions. This project, SkillRank, aims to develop a web-based internship portal integrated with a candidate ranking system using skill-based matching.

The system is developed using the Django web framework, MySQL database, and advanced HTML/CSS for the frontend, making it a full-stack solution suitable for educational and small-to-medium enterprise use.

1.2 Problem Statement

The current internship recruitment process suffers from several limitations:

- Manual screening of resumes is time-consuming and prone to human bias.
- No centralized platform exists for students and employers to interact effectively.
- Lack of automated skill matching and ranking leads to poor candidate-employer fit.
- Students often miss opportunities due to limited visibility.
- Employers receive large numbers of applications without tools to prioritize the most suitable candidates.

These problems result in inefficiency, missed opportunities, and suboptimal hiring outcomes.

1.3 Objectives

General Objective

To design and develop a web-based internship portal with an integrated candidate ranking system to facilitate efficient internship matching between students and employers.

Specific Objectives

- To implement user registration and authentication with role-based access (Intern and Employer).
- To allow users to create and manage role-specific profiles (skills & education for interns, company details for employers).
- To provide a role-based dashboard displaying relevant information and counts.
- To enable employers to post internships (planned for final phase).
- To develop a skill-based candidate ranking mechanism (planned for final phase).

1.4 Scope of the Study
The scope of this project includes:

- User registration, login, and role selection (Intern / Employer)
- Profile management (Intern: skills, education, experience; Employer: company info)
- Role-based dashboard showing basic statistics
- Admin panel for managing users and data
- Basic frontend design using HTML, CSS, and minimal JavaScript

Out of scope (planned for future phases):

- Internship posting form
- Application submission and status tracking
- Full skill matching and ranking algorithm
- Messaging system
- Mobile responsiveness beyond basic level

1.5 Significance of the Study
This project contributes to:

- Reducing manual effort in internship recruitment
- Helping students find better-matched opportunities
- Assisting employers in discovering skilled candidates quickly
- Providing a practical case study of Django web development with custom user models and role-based access
- Serving as a foundation for future enhancements (AI ranking, mobile app, etc.)

1.6 Limitations
- The system is currently web-based and not fully mobile-responsive.
- No real-time notifications or messaging.
- Skill matching and ranking are not yet implemented.
- Limited to two user roles (intern and employer).
- Tested only with sample data (no large-scale deployment yet).

1.7 Organization of the Report
The report is organized as follows:

- Chapter 1: Introduction
- Chapter 2: Literature Review
- Chapter 3: Methodology
- Chapter 4: System Design
- Chapter 5: Implementation (to be added later)
- Chapter 6: Testing & Results (to be added later)
- Chapter 7: Conclusion & Future Work (to be added later)
- References & Appendices

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

In this chapter, we critically examine the existing internship portals, recruitment systems, and candidate ranking methodologies. The aim is to understand the current practices, technologies employed, and limitations of these systems, which will help in defining the scope and objectives of SkillRank. By analyzing both commercial platforms and academic research, this review identifies gaps in automated candidate ranking and internship-specific solutions, highlighting the necessity for a more precise and transparent system.

2.2 Review of Related Works

2.2.1 Internship Portals

LinkedIn Internship Features

LinkedIn, a globally recognized professional networking platform, allows employers to post internships and candidates to apply using their profiles. Its matching algorithm primarily leverages profile data such as skills, education, and work experience. While LinkedIn provides exposure to a large talent pool, its ranking mechanism is not transparent, and it is not exclusively designed for internships. Consequently, candidates often struggle to understand how they are evaluated relative to other applicants (LinkedIn, 2023).

Internshala (India)

Internshala is one of the leading internship portals in India, offering an extensive list of opportunities across multiple categories and locations. It provides basic filtering options, such as location, stipend, and internship category, which assists users in narrowing down opportunities. However, Internshala does not implement automated candidate ranking, leaving the selection process largely dependent on manual employer evaluation (Internshala, 2022). This limitation highlights a significant gap in efficient candidate assessment.

Handshake (USA)

Handshake is a university-oriented platform connecting students with employers, primarily in the United States. It emphasizes employer-student matching through university affiliation, major, and academic performance. While Handshake successfully facilitates connections within regional networks, it lacks a transparent, skill-based ranking mechanism for candidates and is geographically limited, reducing its accessibility for international applicants (Handshake, 2023).

2.2.2 Candidate Ranking Techniques

Recent studies indicate that ranking candidates based on their resumes and profiles has become an important research area in recruitment systems. The following approaches are widely adopted:

Content-Based Filtering

Content-based filtering, which often combines TF-IDF (Term Frequency-Inverse Document Frequency) with cosine similarity, is frequently employed to match candidate resumes with job or internship descriptions. This method analyzes the textual similarity between resumes and internship postings, allowing for automated shortlisting of candidates with relevant skills (Smith et al., 2020). While effective, it relies heavily on keyword matching and may not capture the holistic capabilities of a candidate.

Django-Based Portal Development
Django, a high-level Python web framework, is commonly used for building recruitment and internship portals. It supports custom user models, database management, and the use of signals to automate certain actions such as notifications or ranking triggers (Django documentation, 2023). Combining Django with machine learning models for candidate ranking offers a scalable and efficient approach for modern internship platforms.

2.3 Research Gaps
Based on the review, several gaps are evident in existing systems:

1. Lack of Transparent Ranking – Most commercial platforms do not provide clear visibility into how candidates are ranked, making the selection process opaque.

2. Limited Automation – Manual candidate assessment remains prevalent, particularly in region-specific platforms like Handshake and Internshala.

3. Internship-Specific Focus – Platforms such as LinkedIn cater to general job postings and networking rather than dedicated internship matching.

4. Integration of Advanced ML Models – While machine learning techniques show promise in candidate ranking, their integration into functional web portals is limited.

These gaps underline the need for a dedicated internship portal like SkillRank, which focuses on transparent, skill-based candidate ranking and automated shortlisting using machine learning, while providing an intuitive user interface through a Django-based system.

2.3 Theoretical Framework
The project follows the Software Development Life Cycle (SDLC) – Waterfall model for mid-phase:

Requirement gathering → Design → Implementation → Testing (ongoing)

Matching algorithm will be based on content-based filtering (future phase):

- Extract skills from intern profile and internship requirements
- Compute similarity score using text matching techniques

2.4 Research Gap
- Most platforms lack transparent, skill-focused ranking for internships
- Few open-source projects combine role-based access with candidate ranking
- Limited studies on lightweight Django + MySQL solutions for small-scale internship portals

SkillRank aims to fill this gap by providing a simple, customizable system with future AI ranking.

2.5 Summary
Existing platforms provide basic matching but lack transparent ranking and focus on internships. SkillRank will address this by combining role-based access with planned skill-based ranking.

CHAPTER 3: METHODOLOGY

3.1 Research/Development Method

Agile-inspired iterative development with focus on core modules first:

- Phase 1: Authentication & Role Management
- Phase 2: Profile & Dashboard
- Phase 3: Posting & Application (planned)
- Phase 4: Ranking Algorithm (planned)

3.2 System Requirement Specification (SRS)

Functional Requirements

- User registration with role selection (intern/employer)
- User login/logout
- Role-based profile creation and editing
- Dashboard showing personalized data (counts)
- Admin panel for managing users and data

Non-functional Requirements

- Secure authentication (Django built-in)
- Responsive design (mobile-friendly basic level)
- Data integrity (MySQL transactions)
- Usability (clean UI, intuitive navigation)

3.3 Tools & Technologies Used

- Backend: Django 4.x (Python)
- Database: MySQL
- Frontend: HTML5, CSS3 (custom), minimal JavaScript
- IDE: VS Code
- Version Control: Git
- Server: Django development server

3.4 Feasibility Study

- Technical: Django + MySQL is mature and suitable
- Economic: Open-source tools → zero licensing cost
- Operational: Easy to deploy and maintain
- Schedule: Core modules completed within 4–5 months

3.5 System Architecture

- Client → Browser → Django (views/templates)
- Django → MySQL (models: CustomUser, InternProfile, EmployerProfile)

CHAPTER 4: SYSTEM DESIGN

4.1 Use Case Diagram & Description

Actors

1. Intern – A student seeking internship opportunities.
2. Employer – A company or organization posting internship opportunities.
3. Admin – System administrator managing users and data.

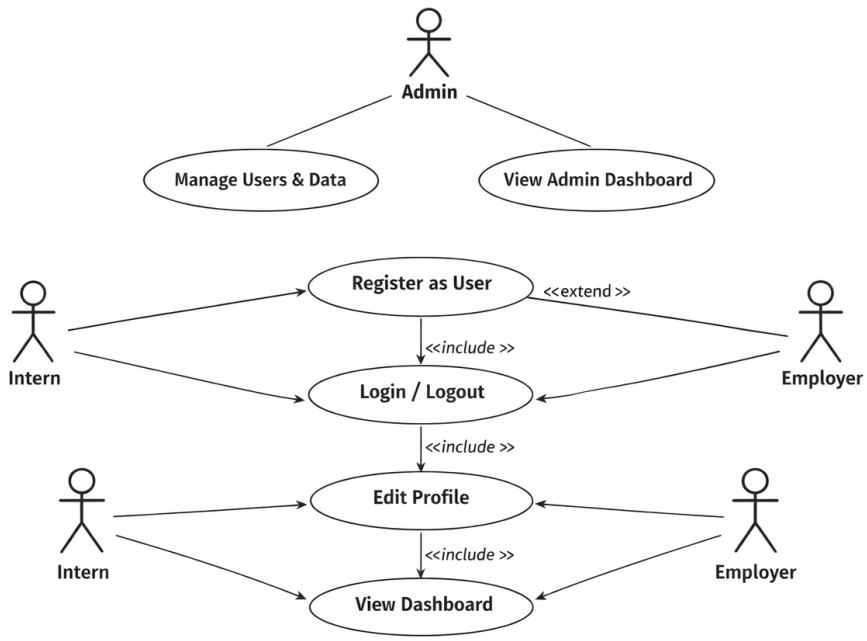
Use Cases

Register as User – Interns and Employers can register with the system and select their role.

1. Login/Logout – Users can authenticate and access the portal.
2. Edit Profile – Users can create or update their role-specific profile.

- Intern: Skills, Education, Experience
- Employer: Company Details

4. View Dashboard – Users see a role-based dashboard with counts and relevant statistics.



4.2 DFD (Level 0, 1, 2)
Level 0- Context Diagram

External Entities:

- Intern – Uses the portal to register, login, and view dashboards.
- Employer – Uses the portal to register, login, and view dashboards.
- Admin – Manages users and monitors data.

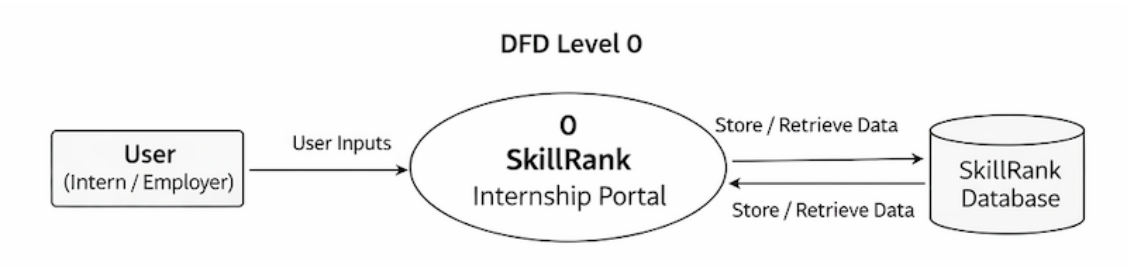
Process:

- SkillRank System – Handles authentication, profile management, dashboards, and admin operations.

Data Stores: None in Level 0.

Data Flows:

- Intern → SkillRank: Registration details, Login info
- Employer → SkillRank: Registration details, Login info
- Admin → SkillRank: Management commands
- SkillRank → Users: Dashboard data, notifications



DFD Level 1

This decomposes the main process into sub-processes:

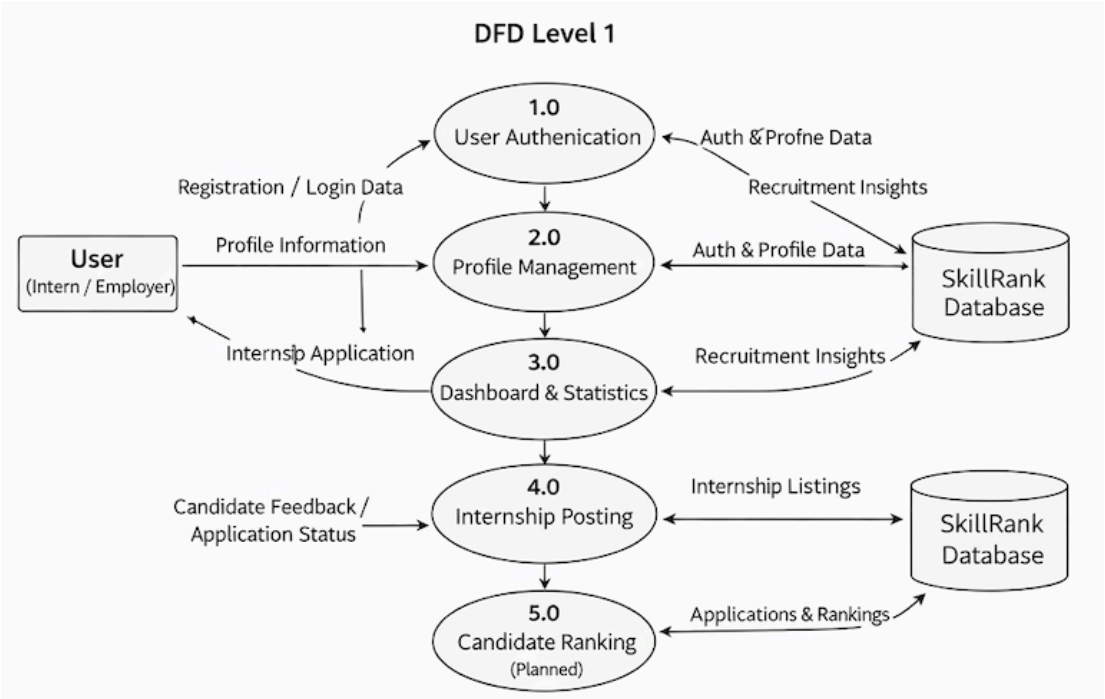
1. User Authentication – Handles login, logout, and role-based access.
2. Profile Management – Allows Interns and Employers to create/edit profiles.
3. Dashboard Management – Displays relevant counts and statistics.
4. Admin Management – Admin can manage users and data.

Data Stores:

- D1: User Data – Stores login credentials and roles.
- D2: Profile Data – Stores intern and employer profile info.

Data Flows:

- Users → User Authentication → User Data
- Users → Profile Management → Profile Data
- Profile Management → Dashboard Management → Users



DFD Level 2

Breakdown of Profile Management and Dashboard for Interns:

Processes:

1. Update Intern Profile – Add/edit skills, education, experience.

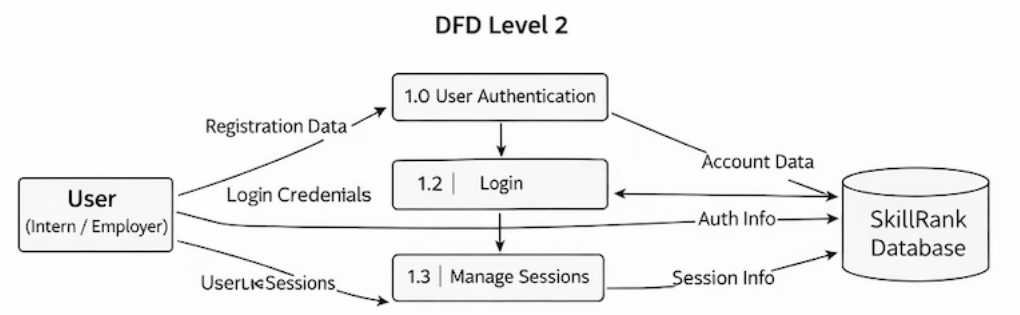
2. Fetch Dashboard Data – Counts of profile completeness, upcoming internships (future phase).

Data Stores:

- D1: Intern Profiles
- D2: System Logs / Stats

Data Flows:

- Intern → Update Profile → Intern Profiles
- Intern Profiles → Fetch Dashboard → Dashboard Info → Intern



4.3 ER Diagram & Database Schema
Entities and Attributes

1. User (superclass / common table)
   - user_id (PK)
   - username
   - email
   - password
   - role (Intern / Employer)
   - date_joined
2. Intern (inherits from User)
   - intern_id (PK, FK → User.user_id)

- skills
- education
- experience

3. Employer (inherits from User)

- employer_id (PK, FK → User.user_id)
- company_name
- company_description
- location

4. Admin (inherits from User)

- admin_id (PK, FK → User.user_id)

5. Internship (planned for future phase)

- internship_id (PK)
- employer_id (FK → Employer.employer_id)
- title
- description
- category
- location
- stipend

6. Application (planned for future phase)

- application_id (PK)
- intern_id (FK → Intern.intern_id)
- internship_id (FK → Internship.internship_id)
- status

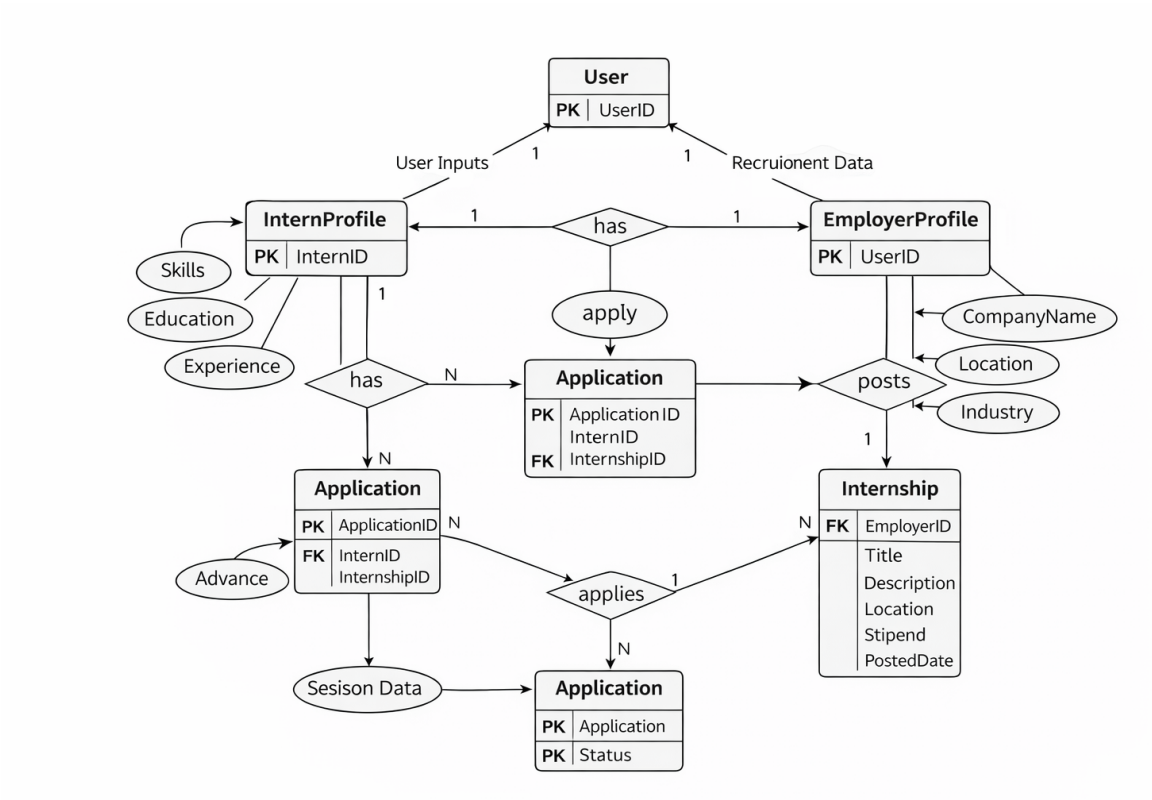
7. Ranking (planned for future phase)

- ranking_id (PK)
- internship_id (FK → Internship.internship_id)
- intern_id (FK → Intern.intern_id)

- score

Relationships

- User → Intern / Employer / Admin: One-to-One (inheritance)
- Employer → Internship: One-to-Many (an employer can post multiple internships)
- Intern → Application → Internship: Many-to-Many via Application
- Intern → Ranking → Internship: Many-to-Many (planned for candidate ranking)



4.4 Class Diagram

CustomUser (AbstractUser)
CustomUser extends Django's AbstractUser and serves as the base user model for both Interns and Employers.

Key Fields:

- id – Primary Key
- username – Unique username
- email – Unique email address
- password – Hashed password
- first_name – User's first name
- last_name – User's last name
- role – Choice field (Intern / Employer / Admin)
- date_joined – Account creation date
- is_active – Boolean (account status)

Relationships:

- One-to-One relationship with InternProfile
- One-to-One relationship with EmployerProfile

A user can only have one profile type depending on their role.

InternProfile (Model)
Stores additional information specific to users registered as Interns.

Key Fields:

- id – Primary Key
- user – OneToOneField → CustomUser
- university – Name of institution
- course – Field of study
- skills – TextField (listed skills)

- resume – FileField (uploaded CV)
- experience_level – Beginner / Intermediate / Advanced
- bio – Short description
- created_at – Profile creation timestamp

Relationships:

- One-to-One with CustomUser
- One InternProfile belongs to exactly one CustomUser
- One Intern can apply to multiple internships (if Internship model exists)

This creates a specialization of CustomUser.

EmployerProfile (Model)
Stores additional information specific to users registered as Employers.

Key Fields:

- id – Primary Key
- user – OneToOneField → CustomUser
- company_name – Organization name
- company_website – URLField
- industry – Company sector
- company_description – TextField
- location – Company address
- verified – Boolean (verification status)
- created_at – Profile creation timestamp

Relationships:

- One-to-One with CustomUser
- One EmployerProfile belongs to exactly one CustomUser
- One Employer can post multiple internships (if Internship model exists)

4.5 Sequence
Sequence for Registration:

1. User → Browser → Enter details → Submit
2. Browser → Django View → Validate → Save CustomUser
3. Django → Signal → Create Profile
4. Django → Redirect to dashboard

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